Aashray Jain

M.Tech Candidate in Artificial Intelligence & Machine Learning

+91-9201876340 | aashrayjain99@gmail.com | github.com/aashrayjain | linkedin.com/in/aashray-jain-b26168119 | aashrayjain.github.io

Research Interests

Trustworthy & Explainable AI (XAI): Post-hoc feature attribution, Grad-CAM interpretability, neuroanatomical biomarker discovery, and class-imbalance formulations in clinical deep learning.
Agentic Systems & Foundation Model Alignment: Model Context Protocol (MCP), multi-agent orchestration, Human-in-the-Loop (HITL) formal safety verification, prompt caching mechanics, and empirical evaluation benchmarking.
Deep Learning Architectures & Medical Imaging: Multi-branch inductive representation fusion, cross-attention mechanisms (Q-MCA), and convolutional benchmarking on axial brain MRI diagnostics.
Distributed Systems & Algorithmic Foundations: Loss function design, context compression algorithms, high-throughput distributed architectures, and streaming pipelines.

Education

- **M.Tech. in Artificial Intelligence & Machine Learning** *2026 – 2028*
  NIMS Marik Institute of Computing, AI, Robotics & Cybernetics, NIMS University Rajasthan First Semester (In Progress)
  – **Coursework:** Advanced Machine Learning, Deep Neural Architectures, Optimization for AI/ML, Probabilistic Modeling & Inference, Advanced Algorithms & Computational Complexity.
- **B.Tech. in Electronics & Communication Engineering** *2017 – 2021*
  Lakshmi Narain College of Technology (LNCT), Bhopal (RGPV) **8.74 / 10.0 CGPA** (Distinction)
  – **Coursework:** Linear Algebra, Probability & Stochastic Processes, Digital Signal Processing, Soft Computing, Computer Networks, Discrete Mathematics.

Research & Deep-Tech Projects

- **Class-Balanced Benchmarking & XAI Clinical DSS (ArihantQNet)** *Oct 2026*
  PyTorch, TorchVision, Grad-CAM (XAI), Scikit-learn, OpenCV, Streamlit Live Workbench | GitHub
  – **Controlled Neuroimaging Benchmarking:** Architected an end-to-end deep learning framework for multi-stage Alzheimer's Disease stratification from 6,400 T1-weighted axial brain MRI scans; conducted controlled benchmarking across ResNet50, DenseNet121, and EfficientNet-B0 with a deterministic 70/15/15 stratified partition (seed = 42).
  – **Mathematical Loss Formulation:** Derived a Normalized Square-Root Inverse-Frequency Weighted Cross-Entropy Loss coupled with AdamW optimization, compressing an extreme 49.78:1 class skew to 7.06:1 and attaining 100% sensitivity on the critically underrepresented Moderate Demented class.
  – **Novel Hybrid Fusion Network (ArihantQNet):** Formulated "ArihantQNet", a tri-branch inductive fusion architecture combining ResNet50 (macro-contextual key), DenseNet121 (fine-boundary query), and EfficientNet-B0 (channel-calibrated value) via 1×1 bottleneck projections, Quantum-Inspired Multi-Head Cross-Attention (Q-MCA), and Squeeze-and-Excitation (SE) recalibration.
  – **Explainable AI (XAI) & Diagnostic Interface:** Synthesized Grad-CAM attribution heatmaps isolating bilateral hippocampal atrophy and ventricular enlargement; deployed a 6-module interactive clinical research workbench for real-time model interrogation and diagnostic reporting.
- **NovaAgent v2: Agentic Orchestration & Formal Safety Verification** *Oct 2026 – Present*
  Model Context Protocol (MCP), Python, FastAPI, PostgreSQL, Prompt-Caching, Docker Novafy OU Core Systems
  – **Protocol-Driven Agent Architecture:** Core contributor to the ground-up rebuild of NovaAgent v2 conversational engine in Python/FastAPI (190+ reviewed commits); architected plug-in tools consuming and serving interfaces over Model Context Protocol (MCP) with dynamic database model catalogs and multi-model fallback chains.
  – **Formal Verification & Human-in-the-Loop Safety:** Designed a transactional safety verification layer for enterprise ERP (MS Dynamics 365 BC); built pre-execution live metadata validation, one-click atomic batch approvals, and non-hallucinatory value-substitution mechanisms ensuring zero schema violations.
  – **Algorithmic Context Optimization:** Formulated request restructuring protocols for provider-side prompt caching, achieving $\sim$99% cached input tokens; engineered dynamic conversation compaction and adaptive fit-to-window budgeting to eliminate context overflow.
  – **Empirical Evaluation Benchmark:** Designed an automated evaluation harness with 70+ deterministic multi-turn scenarios to systematically benchmark prompt drift, reasoning fidelity, and tool precision across containerized PostgreSQL test beds.
- **Multi-Modal Retrieval-Augmented Generation (RAG) & Semantic Search** *Jul 2025*
  Python, LangChain, Hugging Face, FAISS, Streamlit Live Space | Hugging Face
  – Investigated dense semantic retrieval dynamics across diverse open-weights LLMs (DeepSeek, Mixtral, Zephyr) using FAISS indexation; analyzed chunking strategies and semantic embedding geometries to minimize hallucination in technical document querying.
- **ArihantQInvoice: Enterprise Systems Architecture & Intellectual Property** *Apr 2026*
  Node.js, Express.js, SQLite, Sequelize, REST Architecture, OpenAPI Copyright Reg. Govt. of India
  – Architected a full-stack enterprise platform featuring automated document synthesis, inventory state tracking, and 50+ REST endpoints; officially registered software copyright with the Copyright Office, Government of India.

Research & Professional Experience

- **Novafy OU** *Oct 2026 – Present*
  Software Development Intern (AI & Agentic Systems) Remote (Hamburg, Germany)
  - Leading systems engineering and protocol research for NovaAgent v2, standardizing enterprise tool exposure over Model Context Protocol (MCP) and dynamic multi-tenant execution sandboxes.
  - Designed and verified idempotent MS Teams event redelivery handling, replay attack mitigations, structured telemetry, and automated LLM outage detection and alerting pipelines.
  - Implemented isolated integration test suites running against throwaway PostgreSQL test containers, ensuring robust regression prevention across 190+ reviewed pull requests.
- **ArihantQ (AI Research & Innovation Lab)** *Aug 2025 – Present*
  Founder & Independent AI Researcher Remote (India)
  - Directing independent research in LLM reasoning frontiers, agentic orchestration, and cost-performance trade-off modeling; established legal and computational research infrastructure (Udyam Registration, Current Account).
  - Formulated GenAI Cost Optimization Advisor for the Lyzr AI Architect Challenge, engineering multi-source RAG knowledge retrieval and achieving a documented 666% ROI projection (Awarded Cash Prize).
  - Awarded public commendation from MariaDB Foundation and CEO for database innovation and technical open-source contribution.
- **Tata Consultancy Services** *Jul 2021 – Jan 2025*
  Systems Engineer (Distributed Systems & Applied AI/ML) Mumbai & Kochi, India
  - Engineered high-throughput real-time risk analysis dashboards and microservices for a tier-1 Canadian BFSI client, serving 500K+ enterprise users and reducing incident triage latency by 90%.
  - Spearheaded multi-account AWS cloud infrastructure modernization, optimizing architectural resilience and reducing compute/operational costs by 70%.
  - Implemented enterprise-grade authentication frameworks (SAML, AWS Cognito, CyberArk RBAC), cutting security vulnerabilities by 40%.
  - Automated critical enterprise analytical pipelines using Python (Pandas, Flask), Angular, and NX Monorepos, reclaiming 15+ engineering hours per week.
  - Collaborated with TCS Rapid Labs on applied AI/ML proof-of-concepts spanning NLP, Computer Vision, and generative modeling; earlier developed the Asset Procurement module for EquipMax 3.0 during internship (Angular/Ionic).

Academic Honors & Competitive Achievements

- **Top 100 Winner** — Microsoft AI Odyssey Challenge by Microsoft & TCS; 1920-min advanced Azure AI curriculum. *2024*
- **Prize Winner** — Lyzr AI Architect Challenge for algorithmic LLM Cost Optimization Advisor with RAG. *2025*
- **MariaDB Commendation** — Public commendation from MariaDB CEO & Foundation for open-source technical contribution. *2025*
- **National AI Recognition** — **FutureSkills Prime (AI Applications)** by NASSCOM & MeitY, Govt. of India. *2025*
- **Microsoft Learn Honors** — Completed 48 modules, 49 badges, 10 trophies, and 2 Applied Skills credentials in Generative AI. *2024*

Selected Certifications & Technical Credentials

- **Applied AI & Cloud:** **Microsoft**: Applied Skills: GenAI with Azure OpenAI | **Microsoft**: Applied Skills: Document Processing with Azure AI | **AWS**: Cloud Practitioner Essentials
- **Google Cloud AI:** Gemini Multimodality & Multimodal RAG | Explore GenAI with Vertex AI Gemini | Prompt Engineering | GenAI Apps with Streamlit
- **Data Science & ML:** **IBM**: Applied Data Science with Python (Level 2) | **IBM**: Big Data Foundations | **IBM**: Artificial Intelligence Fundamentals
- **Academic Foundations:** **IIT Kanpur (NPTEL)**: Programming in C | **IIT Kharagpur (NPTEL)**: Programming in Java | **HackerRank**: Java (Basic) | **SEBI**: Certified Investor

Technical & Research Toolkit

- **Deep Learning & XAI:** PyTorch, TorchVision, Grad-CAM, Scikit-learn, OpenCV, Hugging Face, FAISS, LangChain, Streamlit
- **Agentic & Backend Systems:** Model Context Protocol (MCP), Python, FastAPI, PostgreSQL, Node.js, Express.js, REST APIs, Docker
- **Cloud & Distributed Computing:** AWS (Lambda, API Gateway, S3, Cognito, DynamoDB), GCP (Vertex AI), Linux, Git, CI/CD
- **Mathematical & Scientific Foundations:** Linear Algebra, Stochastic Optimization, Class-Imbalance Formulations, Signal Processing

Academic Service & Community Leadership

- Active participant in AI/ML hackathons, research seminars, and open-source communities; TCS Blood Donor (Amrita Hospital, Kochi); participant in national technical symposia (Planet Engineers, Convolution) and TEDxLNCTWomen innovation summit.